INCH-POUND

MIL-PRF-1/1105A
27 August 1999
SUPERSEDING
MIL-E-1/1105(SigC)
7 June 1957

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING
TYPE 6564

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Dual T. R. tube (band pass).

Pack in sealed water-vapor-proof bag. If opaque bag is used, the tube type number shall be stamped thereon.

APPLICATION: Used with short-slot hybrid couplers.

DIMENSIONS: See figures 1 and 2.

ABSOLUTE RATINGS:

| Parameter: | Transmitter Po | Open circuit ignitor voltage | Altitude |
|---|---|---|---|
| Unit: | Kw | V dc | ft |
| Maximum: | 200 | --- | 10,000 |
| Minimum: | 1 | -700 | --- |

GENERAL:

The Preproduction sample approval requirements in Signal Corps Drawing SC-A-46600 hereby replace any qualification requirements existing in or otherwise referable to the data in this specification sheet.

(Copies of Signal Corps Drawing SC-A-46600, "Preproduction Sample Approval in Lieu of Qualification Requirements in Specifications," may be obtained from the acquiring activity or as directed by the contracting officer.


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TABLE I. Testing and inspection.

| Inspection | Method | Conditions | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | | | | Min | Max | |
| Qualification | | | | | | |
| Dielectric material | 4101 | | --- | --- | --- | --- |
| Degradation due to vibration | 4021 | G = 10 1/ | --- | --- | --- | --- |
| Low level VSWR | 4473 | F = 8,500 MHz ± 0.1% | δ | --- | 1.2 | --- |
| | | F = 8,800 MHz ± 0.1% | δ | --- | 1.2 | --- |
| | | F = 9,000 MHz ± 0.1% | δ | --- | 1.2 | --- |
| | | F = 9,300 MHz ± 0.1% | δ | --- | 1.2 | --- |
| | | F = 9,600 MHz ± 0.1% | δ | --- | 1.2 | --- |
| Insertion loss | 4416 | F = 8,500 MHz ± 0.1% | --- | --- | 1.0 | dB |
| | | F = 8,800 MHz ± 0.1% | --- | --- | 0.8 | dB |
| | | F = 9,000 MHz ± 0.1% | --- | --- | 0.8 | dB |
| | | F = 9,300 MHz ± 0.1% | --- | --- | 0.8 | dB |
| | | F = 9,600 MHz ± 0.1% | --- | --- | 1.0 | dB |
| | | Ii = 100 µA dc on each ignitor | --- | --- | --- | --- |
| Transmitter - receiver Isolation | 4491 | F = 8,500 MHz ± 0.1% | --- | 15 | --- | dB |
| | | F = 8,800 MHz ± 0.1% | --- | 15 | --- | dB |
| | | F = 9,000 MHz ± 0.1% | --- | 15 | --- | dB |
| | | F = 9,300 MHz ± 0.1% | --- | 15 | --- | dB |
| | | F = 9,600 MHz ± 0.1% | --- | 15 | --- | dB |
| | | Ii = 100 µA dc on each ignitor | --- | --- | --- | --- |
| Ignitor Interaction | 4421 | Ii = 100 µA dc on each ignitor 2/ | --- | --- | 0.1 | dB |
| Ignitor Ignition time | 4401 | Ebb = -700 V dc; R = 4.0 Meg 3/ | t | --- | 5.0 | sec |
| Ignitor voltage drop | 4406 | Ii = 100 µA dc on each ignitor 4/ | --- | -200 | -375 | V dc |
| Flat leakage power | 4452 | F = 9000 MHz ± 0.5 %;<br>po = 40 kw ± 10%;<br>prr = 1,000;<br>tp1 = 1.0 ± 0.1 µs;<br>tp2 = 0.5 ± 0.05 µs;<br>Ii = 100 µA dc on each ignitor | pf | --- | 20 | mw |
| Spike leakage energy | 4452 | F = 9,000 MHz ± 0.5%;<br>po = 40 kw ± 10%;<br>prr = 1,000;<br>tp1 = 1.0 ± 0.1 µs;<br>tp2 = 0.5 ± 0.05 µs;<br>Ii = 100 µA dc on each ignitor | Ws | --- | 0.1 | erg |
| Arc loss | 4488 | F = 9,000 MHz ± 0.5%;<br>po = 40 kw ± 10%;<br>prr = 1,000;<br>tp = 1.0 ± 0.15 µs | --- | --- | 0.8 | dB |
| Temperature cycle | 1027 | | cycles | 1 | --- | --- |
| Temperature cycling life | 1027 | | cycles | 10 | --- | --- |

See footnotes at end of table.



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TABLE I. Testing and inspection - Continued.

| Inspection | Method | Conditions | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | | | | Min | Max | |
| Recovery time (1) | 4471 | F = 9,000 MHz ± 0.5%;<br>po = 200 kw ± 10%;<br>tp = 1.0 µs ± 0.05 µs;<br>prr = 1,000;<br>Ii = 100 µA dc on each ignitor | t | --- | 7.0 | µs |
| Recovery time (2) | 4471 | F = 9,000 MHz ± 0.5%;<br>po = 65 kw ± 10%;<br>tp = 1.0 µs ± 0.05 µs;<br>prr = 1,000;<br>Ii = 100 µA dc on each ignitor | t | --- | 3.0 | µs |
| High level VSWR | 4474 | F = 9,000 MHz ± 0.1%;<br>po = 40 kw ± 10%;<br>tp = 1.0 µs ± 0.05 µs;<br>prr = 1,000;<br>Ii = 100 µA dc on each ignitor | δ | --- | 1.2 | --- |
| Life test | --- | F = 9,000 MHz ± 5%;<br>Ii = 100 µA dc on each ignitor<br>po = 200 kw ± 10%;<br>tp = 1.0 µs ± .05 µs;<br>prr = 1,000;<br>Group D | t | 500 | --- | hours |
| Life test end point: | --- | F = 9,000 MHz ± 0.1% | | | | |
| Recovery time (1) | 4471 | | t | --- | 10 | µs |
| Recovery time (2) | 4471 | | t | --- | 5 | µs |
| Flat leakage power | 4452 | | pf | --- | 20 | mw |
| Spike leakage energy | 4452 | | Ws | --- | .15 | erg |
| Insertion loss | 4116 | | --- | --- | 1.0 | dB |

1/ There shall be no evidence of shorts of the ignitor electrodes as indicated by a short indicator during vibration.

2/ Ignitor interaction is measured with the specified ignitor current adjusted on both ignitor electrodes.

3/ Ignitor ignition time shall be measured separately for each ignitor electrode.

4/ Ignitor voltage drop shall be measured separately for each ignitor electrode.

5/ All measurements that include short-slot hybrids shall be made with gaskets bolted between input and output flanges of tube and mating flanges.



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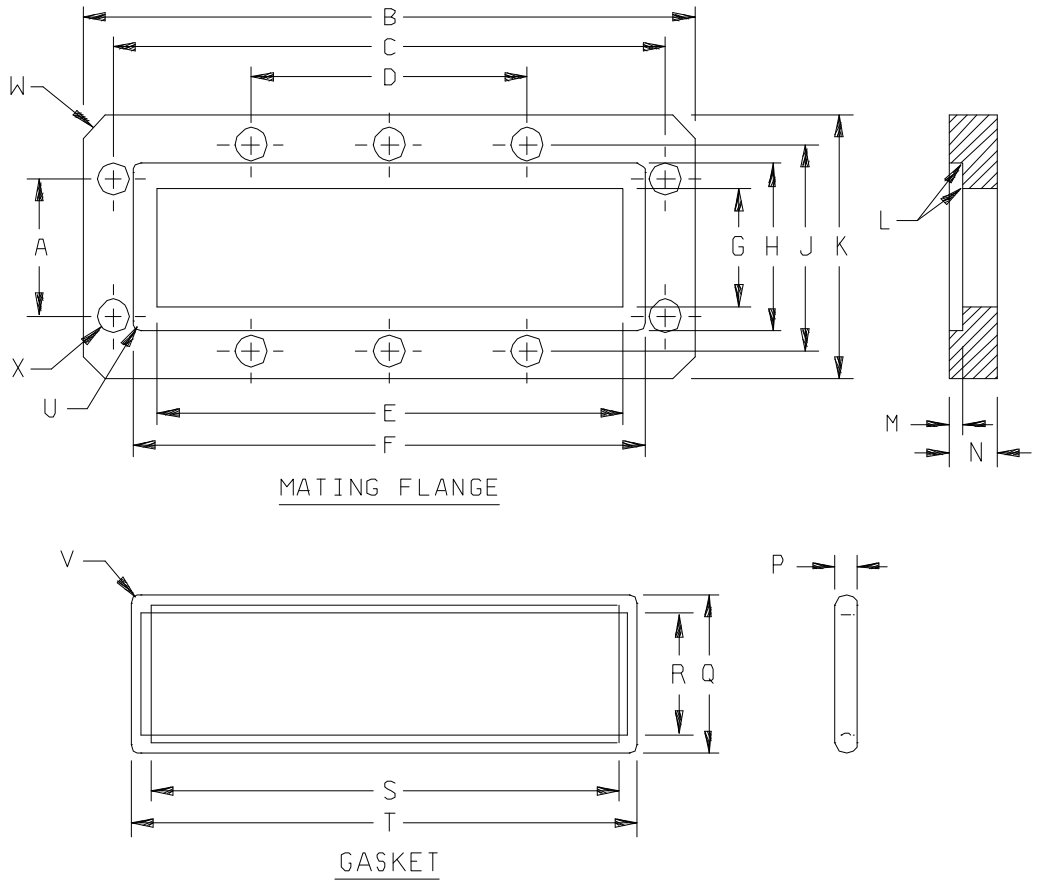


| Ltr | Dimensions | | | | Ltr | Dimensions | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Inches | | Millimeters | | | Inches | | Millimeters | |
| | Min | Max | Min | Max | | Min | Max | Min | Max |
| A | .718 | .726 | 18.24 | 18.44 | M | .067 | .071 | 1.70 | 1.80 |
| B | --- | 3.210 | --- | 81.53 | N | --- | .250 | --- | 6.35 |
| C | 2.884 | 2.900 | 73.25 | 73.66 | P | --- | .110 TYP | --- | 2.79 TYP |
| D | 1.440 | 1.456 | 36.58 | 36.98 | Q | --- | .831 | --- | 21.11 |
| E | 2.439 | 2.442 | 61.95 | 62.03 | R | .630 | --- | 16.00 | --- |
| F | 2.680 | 2.688 | 68.07 | 68.28 | S | 2.444 | --- | 62.08 | --- |
| G | .627 | .630 | 15.93 | 16.00 | T | --- | 2.647 | --- | 67.23 |
| H | .868 | .876 | 22.05 | 22.25 | U | .044 R | .048 R | 1.12 R | 1.22 R |
| J | 1.076 | 1.084 | 27.33 | 27.53 | V | .044 R | .048 R | 1.12 R | 1.22 R |
| K | --- | 1.380 | --- | 35.05 | W | .120 X 45° chamfer or 1.125 R | | 3.00 x 45° chamfer or 28.6 R | |
| L | --- | .005 R | --- | 0.13 R | X | .173 drilled (10) holes | | 4.39 drilled (10) holes | |

FIGURE 1. Mating flange and gasket.



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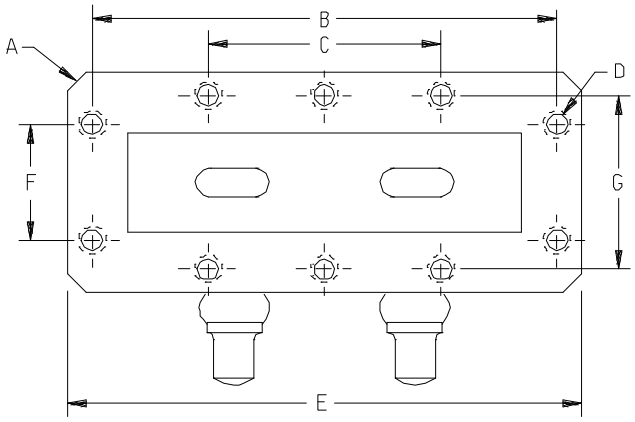
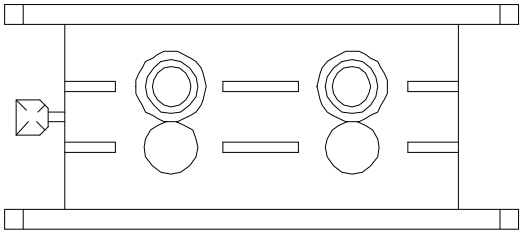
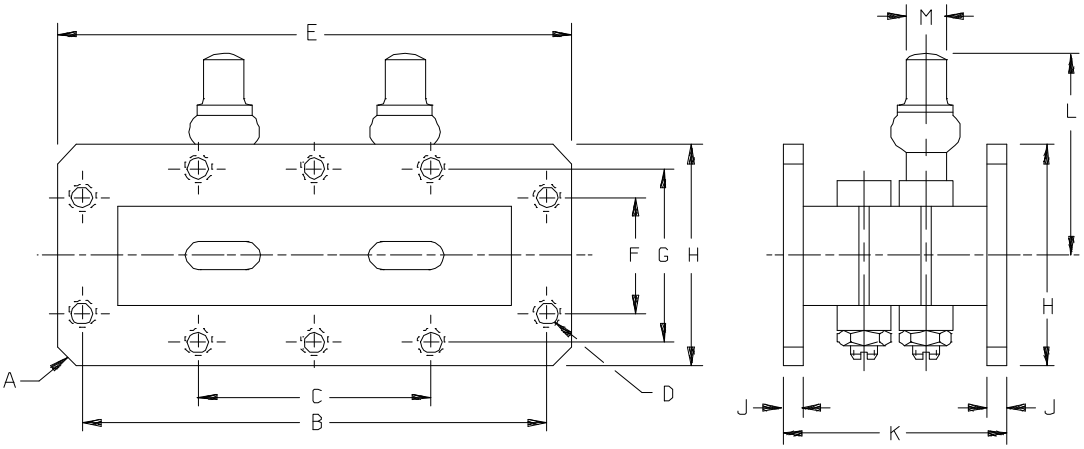



OUTPUT FLANGE



INPUT FLANGE




| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| A | .125 x 45° approx | | 3.20 x 45° approx | |
| B | 2.887 | 2.897 | 73.33 | 73.58 |
| C | 1.443 | 1.453 | 36.65 | 36.91 |
| D | 8-32 Tap (M) holes each figure | | | |
| E | 3.214 | 3.224 | 81.64 | 81.89 |
| F | .717 | .727 | 18.21 | 18.47 |
| G | 1.075 | 1.085 | 27.30 | 27.56 |
| H | 1.370 | 1.380 | 34.80 | 35.05 |
| J | .120 | .130 | 3.05 | 3.30 |
| K | 1.262 | 1.282 | 32.05 | 32.56 |
| L | --- | 1.375 | --- | 34.92 |
| M | .250 Nom | | 6.35 Nom | |

NOTE: Exhaust stem not to extend beyond flange.

FIGURE 2. Input and output flanges.



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Custodians:
Army - CR
Air Force - 11
DLA - CC

Review activities:
Army - MI

Preparing activity:
DLA - CC

(Project 5960-A252)



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